

Sarat Kamandula

Data Engineer | saratkamandula9@gmail.com | +1 (806) 853-7189 | [LinkedIn](#)

SUMMARY

Experienced **Data Engineer** with expertise in building and optimizing cloud-based data platforms. Skilled in designing and orchestrating scalable **ETL/ELT pipelines** using **Databricks (PySpark, Spark SQL, Notebooks, Jobs)** and **Snowflake (SnowPipe, external tables, performance tuning)**, while leveraging **Azure Data Factory** and related services for enterprise-scale integrations. Proficient in **SQL, Python**, and modern cloud ecosystems (**AWS, Azure, GCP**), with a proven track record of enabling self-service BI and developing efficient data lake solutions.

Demonstrated success in optimizing petabyte-scale pipelines to enhance query performance, reduce costs, and deliver faster insights through clustering, caching, and **FinOps** strategies. Adept at data modeling (**Star, Snowflake, and Data Vault**) and ensuring governance, lineage, and data quality using tools such as **Collibra**, Great Expectations, and **Deequ**. Experienced in **Agile/Scrum** environments, collaborating with stakeholders across engineering, analytics, and compliance to deliver data-driven solutions that achieve measurable business-outcomes.

SKILLS

- **Programming & Scripting:** Python, SQL, Java, Scala, Unix Shell Scripting
 - **Databricks:** Notebooks, Jobs orchestration, Delta Lake, Photon acceleration, cluster autoscaling
 - **Snowflake:** SnowPipe, external tables, Iceberg concepts, performance tuning, data warehousing
 - **Data Engineering & Architecture:** Data Modeling (Star, Snowflake, Data Vault, Relational, Dimensional, NoSQL), ETL/ELT pipelines (dbt, Airflow, Glue, ADF), Data Architecture, Data Governance (Collibra, Alation, Apache Atlas), Data Quality & ETL Testing (Great Expectations, Deequ)
 - **Orchestration & Workflow:** Apache Airflow, Prefect, Dagster
 - **Azure Cloud Services:** Data Factory, Synapse, Data Lake, Functions, WebApps, Application Insights, Azure Monitor
 - **Other Cloud Platforms:** AWS (Glue, Lambda, Redshift, Lake Formation, S3, EMR, KMS, IAM, CloudWatch), Azure (Databricks, Synapse, Delta Lake), GCP (BigQuery – exposure)
 - **Streaming & Messaging:** Apache Kafka, AWS Kinesis, Spark Streaming
 - **Data Warehousing & BI:** Snowflake, Redshift, Azure Synapse, PostgreSQL, SQL Server, BI Dashboards (Power BI, Tableau)
 - **APIs & Microservices:** Scalable API development, Docker, Kubernetes, Azure Functions/WebApps
 - **CI/CD & DevOps:** CI/CD pipelines (Git, Jenkins, Docker, Kubernetes, Terraform, GitHub Actions, Azure DevOps)
 - **Monitoring & Observability:** ELK Stack, Datadog, Grafana, Azure Monitor, Cost Management
 - **Security & Compliance:** IAM, RBAC, Encryption, GDPR, HIPAA, SOX
 - **Cost Optimization & FinOps:** Cloud cost management, performance tuning, resource optimization
 - **Methodologies:** Agile, Scrum, SDLC
-

EDUCATION

- Master's in Computer Science, Texas Tech University, TX
 - AWS Certified Solutions Architect - Associate
-

PROFESSIONAL EXPERIENCE

Data Engineer | TD Bank

Aug 2024 – Current

- Designed **AWS Glue, Lambda, and S3** pipelines for petabyte-scale datasets, improving throughput by 30% for financial processing.
- Built Apache **Airflow DAGs**, reducing pipeline runtime by 40%, and implemented **Snowflake** data models for compliance and reporting.
- Developed **Databricks** Jobs and Notebooks using **PySpark** and **Spark SQL** for scalable large-scale data processing workflows.
- Configured **Databricks** clusters with autoscaling and **Photon** acceleration, increasing performance by 30% in big data workloads.
- Implemented **SnowPipe** and external tables for real-time ingestion in **Snowflake**, reducing reporting latency significantly.
- Built ETL/ELT pipelines using **Azure Data Factory**, integrating monitoring with Application Insights and **Azure Monitor**.
- Designed **Snowflake** and **Redshift** database models with architects, aligning schemas to business rules and compliance needs.
- Established governance and lineage using **Collibra** and **Lake Formation**, automating deployments with **Jenkins**, **GitHub Actions**, and **Terraform**.
- Developed dashboards in **ELK** and **Datadog** for real-time monitoring, applying **FinOps** to reduce processing costs by 20%.
- Collaborated with business and technical teams in **Agile/Scrum**, ensuring stakeholder alignment on deliverables and requirements.

- Contributed to petabyte-scale pipelines powering regulatory compliance reporting and enabling enterprise-wide BI analytics for financial datasets.

Data Engineer | HealthifyMe

Oct 2020 – Jul 2023

- Built data pipelines with **AWS Redshift** and **Glue** to automate **ingestion** and **transformation**, improving query performance by **40%**.
- **Developed REST APIs** for pipeline integration and **containerized** with **Docker** for scalable deployment.
- Used **Kafka** for ingesting real-time user activity data, enabling personalized insights and recommendations.
- Leveraged **MongoDB** for semi-structured storage and optimized retrieval, cutting processing time by **30%**.
- Orchestrated **pipelines** using **Airflow**, enhancing **data reliability** and observability across **ingestion layers**.
- Designed and enforced **data quality checks** and **ETL testing using Great Expectations**, reducing downstream data issues by **25%**, applied data security best practices with **IAM** and **KMS** for sensitive user data.
- Containerized **pipeline workloads** with **Docker** and automated deployments with **CI/CD**, ensuring consistent environments, partnered with product managers and analysts in an **Agile/Scrum** framework to deliver features iteratively.
- Delivered **real-time analytics pipelines** that improved **user personalization** and reduced downstream **data quality issues** by **25%**, directly enhancing customer engagement.

Data Engineer | Data Capital

Aug 2019 – Sep 2020

- Designed and developed **Azure-native ETL pipelines** using **Data Factory** and **Databricks** (with **Delta Lake**), reducing ETL time by **30%**, modeled structured/unstructured data into **Azure Synapse** and **Snowflake** to enable **BI** reporting and analytics.
- Designed and deployed **Delta Lake pipelines on Databricks**, leveraging **Iceberg** concepts for scalable **data lake management**.
- Built microservices using **Azure Functions** and **WebApps** to support **API-based** data ingestion and monitoring with **Application Insights**.
- Developed **ETL/ELT workflows** across **Azure Data Factory** and **Databricks** for **structured** and **unstructured** data integration.
- Implemented **big data** workflows with **Hadoop** and **Spark**, improving large-scale processing efficiency.
- Developed **dbt** models for transformations, ensuring transparency, modularity, and **version control** in **SQL workflows**.
- Built executive dashboards in **Power BI**, delivering actionable insights to business stakeholders.
- Automated deployments via **Azure DevOps** pipelines and **Kubernetes**, reducing release cycles by **50%**, incorporated compliance, security, and audit readiness controls for financial reporting, participated in **Scrum** sprints.
- Built **Azure-native pipelines and dashboards** that reduced **ETL runtime** by **30%**, improved **financial reporting accuracy**, and accelerated **executive decision-making**.